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3)

$|AV| = \frac{V_o}{V_{in}}$

$G_V = 20 \log_{10} |AV|$

$A_V = \frac{\Sigma (V_{in} \cdot V_o)}{\Sigma V_{in}^2}$

$\Sigma (V_{in} \cdot V_o) = 0.04 \times 1.76 + 0.06 \times 2.68 + 0.08 \times 3.56 + 0.10 \times 4.32$

$= 0.9480$

$= \frac{0.9480}{0.0216}$

$= 43.9$

$\Sigma (V_{in}^2) = 0.04^2 + 0.06^2 + 0.08^2 + 0.10$

$= 0.0216$

$G_V = 20 \log 43.9$

$= 32.8$ dB

測定 $\Sigma (V_{in} \cdot V_o) = 0.9596$

$\Sigma (V_{in})^2 = 0.0224$

$A_V = 43.618816$

$G_V = 32.79335081$

$\fallingdotseq 32.8$

目標値 $32.8 - 1 = 31.8$

32.0 120mV

31.6 140mV

$V_{in} = 120 + \frac{20 \times (32.0 - 31.8)}{(32.0 - 31.6)}$

$\fallingdotseq 129 \times 10^{-3}$

実測レポート

$h_{ie} = 29.4$

$h_{fe} = 170$

$R_{ac} = \frac{R_C R_L}{(R_C + R_L)}$

$= \frac{510 \times 510\Omega}{(510 + 510\Omega)}$

$= 463.6\Omega$

$R_V = -\frac{h_{fe} R_{ac}}{h_{ie}}$

$R_V = \frac{-h_{fe} \cdot R_{ac}}{h_{ie}}$

$= -\frac{29.4 \cdot 463.6}{170}$

$= -26.8$

$G_V = 20 \log |26.8|$

$\fallingdotseq 28.8$

理論値 $A_V = -26.8$

$G_V = 28.6$

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$\frac{43.9}{26.8} = 1.68$

$\frac{(43.9 - 26.8)}{26.8} \times 100 = 63.8$

$32.8 - 28.6 = 4.2$

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$A_V = \frac{V_C}{V_{in}}$.

$G_V = 20 \log A_V$.

最大値　34.3dB ($V_o = 2.08V$, $|A_V| = 52.0$)

$G_V = 34.3 - 3$

$= 31.3$.

$|A_V| = \frac{52.0}{\sqrt{2}}$

$= 36.8$.

$V_o = \frac{2.08}{\sqrt{2}}$

$= 1.47$.

$\log_{10} f_L = \log_{10} 100 + \frac{(31.3 - 28.1)}{(32.5 - 38.1)} (\log_{10} 200 - \log_{10} 100)$

$= 2.219$.

$f_L = 10^{2.219}$

$= 166 Hz$.

166Hz